

API Endpoints Documentation

This document lists all API endpoints for the backend service. Use this as a reference for Postman testing.

Server Information

- Base URL: `http://localhost:5000`

General Chat Endpoints

Ask a General Question

- **Endpoint:** `/ask`
- **Method:** POST
- **Description:** Ask a general question. If college-related, it will try to find context from college data.
- **Request Body:**

```
{
  "text": "Tell me about the college campus"
}
```

- **Response:**

```
{
  "answer": "The response from the AI..."
}
```

PDF Management Endpoints

Upload PDF

- **Endpoint:** `/upload-pdf`
- **Method:** POST
- **Description:** Upload a PDF file for processing
- **Request:** Form data with key `pdf` containing the PDF file
- **Response:**

```
{
  "pdfId": "64f89a2e1234567890abcdef",
  "message": "✅ PDF uploaded and embedded."
}
```

Ask Question about PDF

- **Endpoint:** `/ask-pdf`
- **Method:** POST
- **Description:** Ask a question about a previously uploaded PDF
- **Request Body:**

```
{
  "pdfId": "64f89a2e1234567890abcdef",
  "question": "What is the main topic of this document?"
}
```

- **Response:**

```
{
  "answer": "The response from the AI based on the PDF content..."
}
```

Get Chat History for PDF

- **Endpoint:** `/chat-history/:pdfId`
- **Method:** GET
- **Description:** Get the chat history for a specific PDF
- **Parameters:**
 - `pdfId`: The ID of the PDF document
- **Response:**

```
{
  "history": [
    {
      "_id": "64f89a2e1234567890abcdef",
      "pdfId": "64f89a2e1234567890abcdef",
      "pdfTitle": "Document.pdf",
      "question": "What is this document about?",
      "answer": "This document is about...",
      "createdAt": "2023-09-06T12:34:56.789Z"
    }
  ]
}
```

Get All Chats

- **Endpoint:** `/all-chats`
- **Method:** GET
- **Description:** Get all chats grouped by PDF
- **Response:**

```
[
  {
    "_id": {
      "pdfId": "64f89a2e1234567890abcdef",
      "pdfTitle": "Document.pdf"
    },
    "chats": [
      {
        "question": "What is this document about?",
        "answer": "This document is about...",
        "createdAt": "2023-09-06T12:34:56.789Z"
      }
    ]
  }
]
```

Save Chat

- **Endpoint:** `/save-chat`
- **Method:** POST
- **Description:** Save a chat interaction for a PDF
- **Request Body:**

```
{
  "pdfId": "64f89a2e1234567890abcdef",
  "question": "What are the main points?",
  "answer": "The main points are..."
}
```

- **Response:**

```
{
  "message": "Chat saved successfully"
}
```

Alternative PDF Routes (Currently Not Used in Main Server)

These routes are defined in `routes/pdfRoutes.js` but are not currently integrated into the main server:

Upload PDF (Alternative)

- **Endpoint:** `/pdf/upload`
- **Method:** POST
- **Description:** Upload a PDF file (using LangChain implementation)
- **Request:** Form data with key `pdf` containing the PDF file
- **Response:**

```
{
  "success": true,
  "filename": "1630948123456-document.pdf",
  "chunks": 15
}
```

Ask PDF Question (Alternative)

- **Endpoint:** `/pdf/ask`
- **Method:** POST
- **Description:** Ask a question about a PDF (using in-memory chunks storage)
- **Request Body:**

```
{
  "filename": "1630948123456-document.pdf",
  "question": "What is this document about?"
}
```

- **Response:**

```
{
  "response": "This document is about..."
}
```

List PDFs (Alternative)

- **Endpoint:** `/pdf/list`
- **Method:** GET
- **Description:** Get a list of all uploaded PDFs
- **Response:**

```
{
  "files": [
    {
      "filename": "1630948123456-document.pdf",
      "chunks": 15
    }
  ]
}
```

Get PDF Chunks (Alternative)

- **Endpoint:** `/pdf/chunks/:filename`

- **Method:** GET
- **Description:** Get all chunks from a specific PDF
- **Parameters:**
 - **filename:** The filename of the PDF
- **Response:**

```
{  
  "filename": "1630948123456-document.pdf",  
  "chunks": ["chunk 1 text", "chunk 2 text", "..."]  
}
```

Notes for Testing

1. The main implementation is in `server.js` which uses MongoDB for storage and Python FAISS for vector search.
2. The alternative implementation is in `routes/pdfRoutes.js` and `controllers/pdfController.js` but is not currently connected to the main server.
3. For PDF uploads, use multipart/form-data with the key 'pdf'.
4. Remember to store the returned pdfid from uploads for subsequent calls to /ask-pdf and /chat-history.